

Department of Electrical and Computer Engineering
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ECE 552 Introduction to Computer Architecture
Homework 7 (Due 11/03)

1. Consider the following trace of memory read references of **8-bit memory addresses** (given in decimal). Memory is byte-addressable.
21, 166, 201, 143, 61, 166, 62, 133, 111, 143, 144, 61
- (a) You are asked to improve the cache design for the above trace. There are three **direct-mapped** cache designs possible, all with a **32-byte total capacity**: C1 has 4-byte blocks, C2 has 8-byte blocks and C3 has 16-byte blocks. First, fill in the table below. Assume each cache is initially empty. Then specify which cache design(s) yields the **lowest miss rate** for this trace of memory references.

Answer: (blanks are Miss)

Trace	Binary address	Tag	C1		C2		C3	
			Index	H/M	Index	H/M	Index	H/M
21	00010101	000	101		10		1	
166	10100110	101	001		00		0	
201	11001001	110	010		01		1	
143	10001111	100	011		01		0	
61	00111101	001	111		11		1	
166	10100110	101	001	H	00	H	0	
62	00111110	001	111	H	11	H	1	H
133	10000101	100	001		00		0	
111	01101111	011	011		01		0	
143	10001111	100	011		01		0	
144	10010000	100	100		10		1	
61	00111101	001	111	H	11	H	1	

C1: miss rate = 9/12 = 75%

C2: miss rate = 9/12 = 75%

C3: miss rate = 11/12 = 91.7%

Thus C1 and C2 have the lowest miss rate.

- (b) Suppose the above trace is applied to a **3-way set-associative** cache with 8-byte blocks and a total capacity of 96B. Assume **LRU replacement**. For each reference, specify the set index bits, tag contents **after** the access, and hit/miss. The cache is initially empty, and fill in empty tags from left to right.

Answer: Set index = 2 bits. Byte offset = 3 bits. Tag = $8 - 2 - 3 = 3$ bits.

Trace	Binary address	Index	Tag #1	Tag #2	Tag #3	H/M
21	00010101	10	000			
166	10100110	00	101			
201	11001001	01	110			
143	10001111	01	110	100		
61	00111101	11	101	001		
166	10100110	00	101			H
62	00111110	11	101	001		H
133	10000101	00	101	100		
111	01101111	01	110	100	011	
143	10001111	01	110	100	011	H
144	10010000	10	000	100		
61	00111101	11	101	001		H

- (c) Suppose the above trace is applied to a **fully-associative** cache with 8-byte blocks and a total capacity of 32B. Assume **LRU replacement**. For each reference, specify the tag contents **after** the access, and if it is a hit or a miss. The cache is initially empty, and fill in empty tags left to right.

Answer: Byte offset = 3 bits. Tag = 8 – 3 = 5 bits.

Trace	Binary address	Tag #1	Tag #2	Tag #3	Tag #4	H/M
21	00010101	00010				
166	10100110	00010	10100			
201	11001001	00010	10100	11001		
143	10001111	00010	10100	11001	10001	
61	00111101	00111	10100	11001	10001	
166	10100110	00111	10100	11001	10001	H
62	00111110	00111	10100	11001	10001	H
133	10000101	00111	10100	10000	10001	
111	01101111	00111	10100	10000	01101	
143	10001111	00111	10001	10000	01101	
144	10010000	10010	10001	10000	01101	
61	00111101	10010	10001	00111	01101	

2. Assume a processor with a one-level cache hierarchy: L1-I cache, L1-D cache and memory. The **base CPI** (assuming all cache accesses are hits) is 1.

Consider the following program and cache behavior:

- Loads per 1000 instructions: 250
- Stores per 1000 instructions: 100
- L1-I miss rate: 0.3%
- L1-D miss rate: 2%

What is the **maximum cache miss delay** (in cycles) needed to achieve an average **CPI of 2** for this program? Assume that loads and stores incur the same miss delay, and the L1-I and L1-D caches incur the same miss delay.

Answer:

Let **M** be the miss delay.

- **L1-D load miss penalty (CPI) = $250/1000 \times 2\% \times M$**
- **L1-D store miss penalty (CPI) = $100/1000 \times 2\% \times M$**
- **L1-I miss penalty (CPI) = $0.3\% \times M$**

$$\text{CPI} = 1 + (0.25 \times 2\% + 0.1 \times 2\%) \times M + 0.3\% \times M \leq 2$$

Thus **$M \leq 100$ cycles.**